

CHAPTER 4

INTRODUCTION TO NATURAL FREQUENCY TESTING AND INSTRUMENTATION

4.0 INTRODUCTION

Every mechanical system has a series of natural frequencies, each of which has its own damping characteristics. These natural frequencies will lie "dormant" in a system until they are excited by some external influence or forcing function. The vibration can be greatly amplified if a forcing function, such as unbalance or a blade pass frequency (#blades X RPM) happens to be within the range of a natural frequency. When such forcing frequencies coincide with (or closely approach) a natural frequency, this is the condition known as resonance. In some machines (such as reciprocating or diesel engines) which are subject to impact, several such natural frequencies can be excited for reasons which will be explained later in this paper. The effects will be particularly pronounced if the frequency is sustained over a significant time duration within a system's resonant range. Some of the effects of this amplified vibration can range from premature wear and excessive maintenance to fatigue failure or complete structural collapse (Ref. 1).

Solving vibration problems due to resonance can increase uptime and decrease maintenance, replacement, and operating costs. It is a two-part process. First, the resonant frequencies need to be identified and compared to the operating speeds of the equipment. Second, to resolve the problem, a choice has to be made between changing the operating speed or changing the resonant frequencies (or both) (Ref. 1).

This chapter will introduce the analyst to various techniques and instrumentation that can be used to test and measure natural frequencies. In particular, impact/impulse testing methods will be covered, along with coastdown and runup tests as well as Bode' plot and Polar plot acquisition and interpretation.

4.1 DIFFERENCE BETWEEN NATURAL FREQUENCY, RESONANCE AND CRITICAL SPEED

The terms "natural frequency", "resonance" and "critical speed" are often used interchangeably, even by experienced vibration analysts. However, there are distinctions between each of these three terms. For example, a machine may be operating at a resonant frequency, but this may or may not be a critical speed. It is hoped the following definitions will help clarify the differences between each of these important terms.

a. Natural Frequency:

Reference 2 defines a natural frequency as "The frequency of free vibration of a system. The frequency at which an undamped system with a single degree of freedom will oscillate upon momentary displacement from its rest position." In addition, for a multiple degree-of-freedom system, the natural frequencies are the frequencies of the normal modes of vibration. All machines and all structures have a number of natural frequencies. If forced to vibrate at one or more of these natural frequencies, dynamic stresses of 10 to 100X higher than those are induced compared to those which would be generated if these same forces were input at other frequencies lower or higher than these natural frequencies. However, just because a machine may

have many natural frequencies is not necessarily a problem. It only becomes a problem if they are excited (see "resonance" below). For example, when we perform impulse tests, we are not looking for resonant frequencies, but for natural frequencies which the machine must avoid so that it may not encounter resonance.

b. Resonance:

Resonance is the condition which occurs when such forcing frequencies do in fact coincide with one or more natural frequencies. These may be a natural frequencies of the rotor, but often can be a natural frequency of the support frame, foundation or even of drive belts. The piping on compressors can be a problem if they resonate. Forcing frequencies include those from sources such as unbalance, misalignment, looseness, bearing defects, gear defects, belt wear, etc. Therefore, it is important to point out resonances can be encountered *not only at 1X RPM*, but at series of frequencies including 1X RPM, 2X RPM, 3X RPM, etc. Reference 2 defines resonance as "the condition of vibration amplitude and phase change response caused by a corresponding system sensitivity to a particular forcing frequency. A resonance is typically identified by a substantial amplitude increase, and related phase shift."

c. Critical Speed:

Reference 3 points out that "*critical speeds are a special case of resonance in which the vibrating forces are caused by the rotation of the rotor*". He also adds that "*critical speed testing is often more complicated than resonance testing because the natural frequencies encountered are functions of stiffness and mass which may be dependent on machine speed*". For example, *in the case of a machine with an extended overhung rotor or in the case of general sleeve bearing machines, critical speeds can change their frequency location due to changes in stiffness and/or gyroscopic motion (unlike resonant frequencies for frames, foundations and rolling element bearing machines which have fixed natural frequencies independent of operating speed)*. Similarly, Reference 1 defines critical speeds as "in general, any rotating speed which is associated with high vibration amplitude. Often, the rotor speeds which correspond to natural frequencies of the system." Generally, good design practices mandate that a machine should be designed not to operate within 20% of a critical speed (*not 20% of any natural frequency*).

The point made in the last sentence of the preceding paragraph is that it is practically impossible to operate at least 20% away from any natural frequency (of the rotor, support frame, foundation and attached structure). And, any structure has six (6) different sets of natural frequencies including those in the horizontal, vertical and axial directions, as well as "rocking modes" in each of 3 directions. Thus, the key will be to identify the locations of all resonances being excited, how severe the resonant vibration is, and what is the duration of time to which the rotating or stationary structure is subjected to resonance. Finally, if the machines and structures are fabricated from steel, cast iron, aluminum or any other lightly damped material, it may only be necessary to remain approximately 10% away from resonance in order to avoid resonant amplitude amplification. This step alone can easily reduce vibration from 10 to as much as 30 times.

Figure 1 helps illustrate what happens when a machine goes into resonance. It shows a graph of magnification factor (Q) on the vertical axis versus frequency ratio (f/f_n) on the horizontal axis on the upper graph, while plotting phase lag (degrees) versus the same frequency ratio in the lower graph. Note that the frequency ratio is the ratio of forcing frequency (f) to undamped natural frequency (f_n). This forcing frequency can be caused by many sources and can occur at many different frequencies. For example, it can be caused by unbalance at 1X RPM; or by misalignment possibly at 2X RPM; or by a gear mesh frequency at the number of gear teeth X RPM; or by blade pass frequency (BPF) at the number of blades (or vanes) X RPM; or even excited by a rolling element bearing defect frequency. Figure 1 shows that the problem occurs

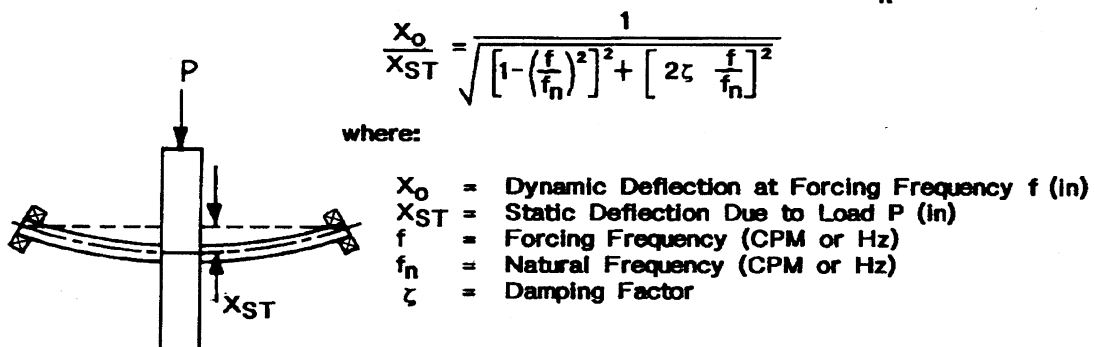
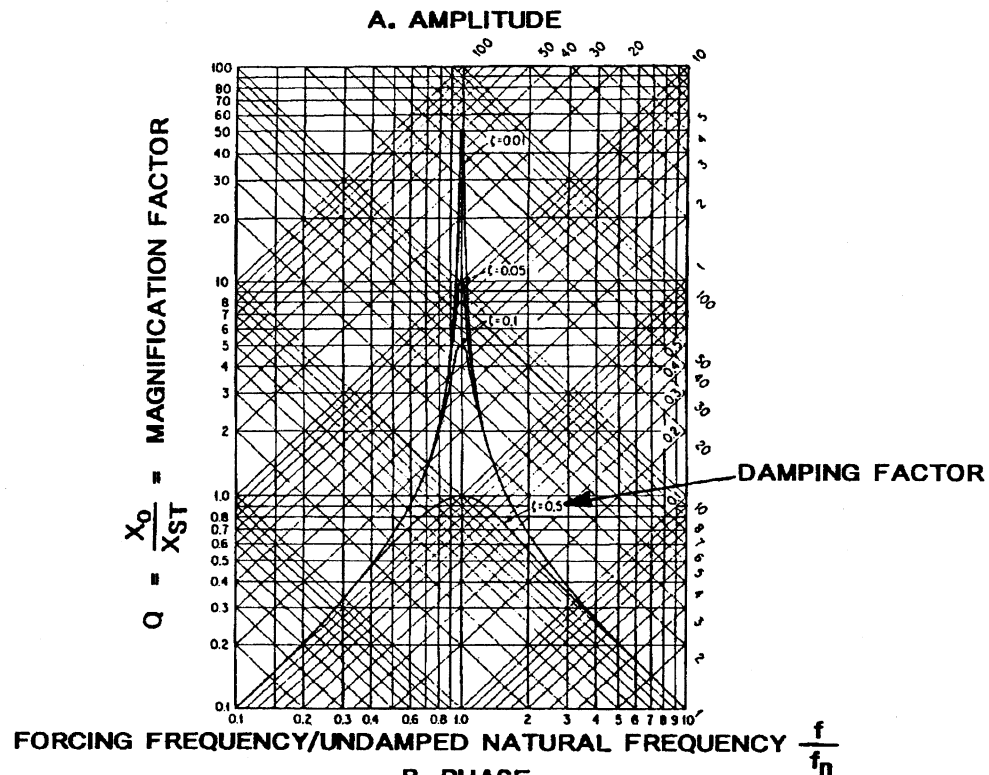
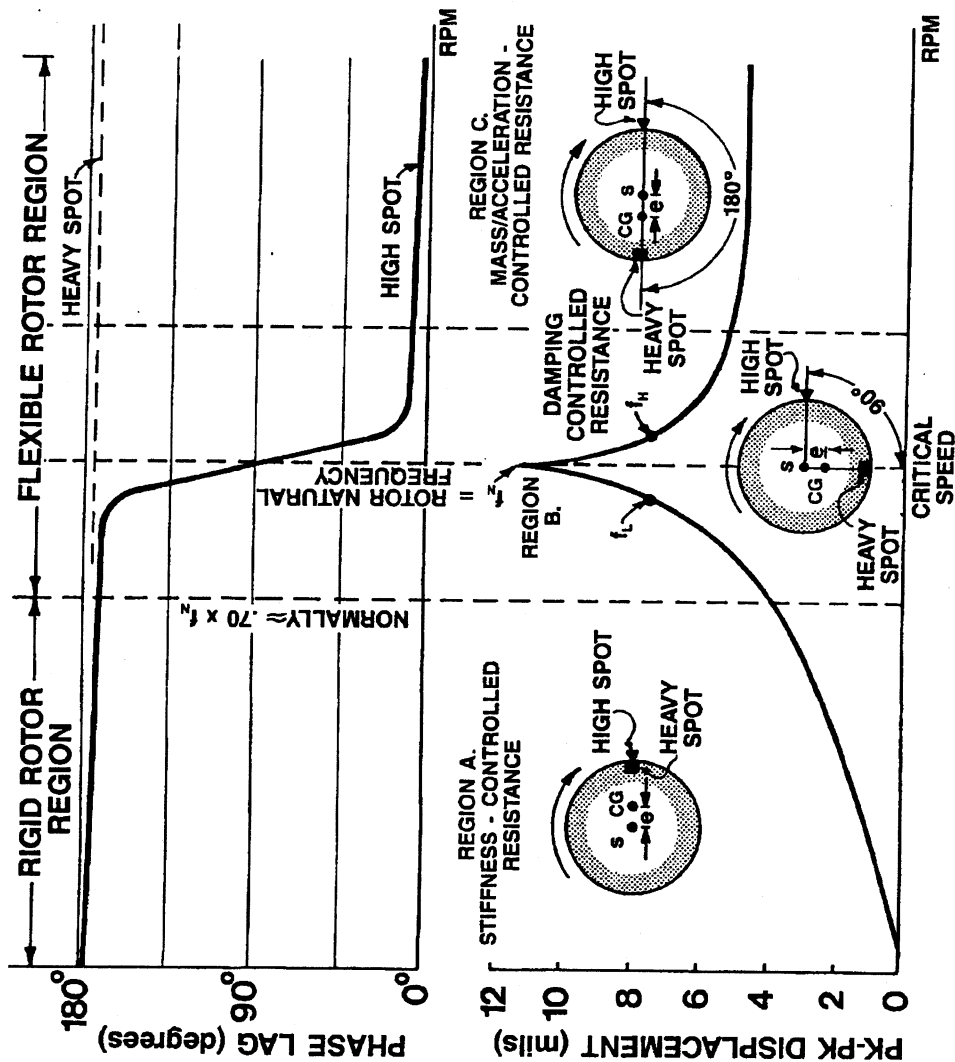


FIGURE 1
RESONANT AMPLIFICATION CURVE & PHASE CHANGE AT RESONANCE



$$\frac{X_d}{X_{st}} = \frac{1}{\sqrt{\left[1 - \left(\frac{f_i}{f_n}\right)^2\right]^2 + \left[2d \frac{f_i}{f_n}\right]^2}}$$

When Forcing Freq. = Natl. Freq. ($f_i = f_n$):

$$\frac{X_d}{X_{st}} = \frac{1}{2d} = Q = \frac{f_n}{(f_n - f_i)}$$

WHERE:

- X_d = Dynamic Deflection at Frequency f_i (mils)
- X_{st} = Static Deflection (mils)
- f_i = Forcing Frequency (CPM)
- f_n = Natural Frequency (CPM)
- d = Damping Factor (lb-sec/in)
- Q = Resonant Amplification Factor
- f_h = .707 A_n = Upper Half-Power Freq. (CPM)
- f_l = .707 A_n = Lower Half-Power Freq. (CPM)
- A_n = Amplitude of Natural Frequency (mils)

LEGEND

- CG = Rotor Center of Gravity
- S = Shaft Geometric Centerline
- e = Eccentricity (in.)
- Heavy Spot = Actual Unbalance Location
- High Spot = Point of Maximum Rotor Deflection

FIGURE 2
CHANGE OF VIBRATION DISPLACEMENT AND PHASE LAG WITH RPM ABOVE, BELOW,
AND AT ROTOR RESONANCE

when such a forcing frequency (f) happens to equal the system natural frequency (f_n) - that is, when $f/f_n = 1.0$. When this occurs, Figure 1 shows that vibration can be amplified many times higher, often on the order of 10 or even 50 times higher vibration, depending on the amount of damping within the system (*a lightly damped system will suffer a tremendous increase when vibrating at a natural frequency*).

Therefore, *this magnification factor (Q) relates how much vibration is amplified when a machine passes through a natural frequency*. If a structure is lightly damped (damping factor = .001), Figure 1 shows that vibration can be amplified approximately *50 times higher*. In the case of most machine component materials of construction, these normally have damping factors (ζ) ranging from approximately .001 up to as high as .050. Looking at Figure 1, this means that when passing through resonance, most machines might experience an increase from 10 times up to 50 times higher vibration if the machine is allowed to dwell on the natural frequency. Therefore, such vibration can easily lead to premature, or even *catastrophic machine failure*.

Thus, resonant vibration problems can be very severe, and within the last few years, have been found to be more commonplace than ever before. Thus, the remainder of this chapter will be devoted to introducing the analyst on how to conduct some rather simple tests for natural frequencies should he suspect them to be present within his machinery, support frame, foundation, or even attached piping and building structure. Later, the "ANALYSIS III" seminar will go into much greater depth on how to not only detect natural frequency problems (using both single-channel and multi-channel analyzers), but also on how to correct them using proven, cost effective solutions.

4.2 CHANGE IN MODE SHAPE WITH HIGHER NATURAL FREQUENCIES

A "**Mode Shape**" is the resultant deflected shape of a structure when forcing frequencies coincide with natural frequencies of the structure (i.e., condition of "resonance"). Figure 3 shows how a rotor simply supported between bearings deflects when it passes through its first, second and third critical speeds. Note the change in mode shape as the rotor passes through its first, second and third critical speeds. Drawing A shows that the maximum deflection will be at the center of the roll when it passes through its first critical. At this point, it will have minimum displacement at each end where the bearings are located.

Drawing B of Figure 3 shows the *distinctly different mode shape as the roll enters its second critical*. Note here that the center point (mid-span) now is a point of minimum deflection (displacement). These points of minimum deflection are known as "**Nodal Points**". A nodal point (also called a "node") are often at a point of minimum absolute shaft displacement. *Motion immediately on either side of a node is approximately 180° out-of-phase*. It is important to point out that nodal points may readily change location along the shaft axis due to changes in residual unbalance or other forcing function, or due to a change in restraint such as an increased bearing clearance.

Drawing C of Figure 3 shows the mode shape when the roll enters its third critical speed. Here, node points will again be on either end, but *two other node points will now occur at approximately one-third and two-thirds of the bearing center distance*. There will now be three *antinodes*, or points of maximum deflection each of which will occur between the bearings.

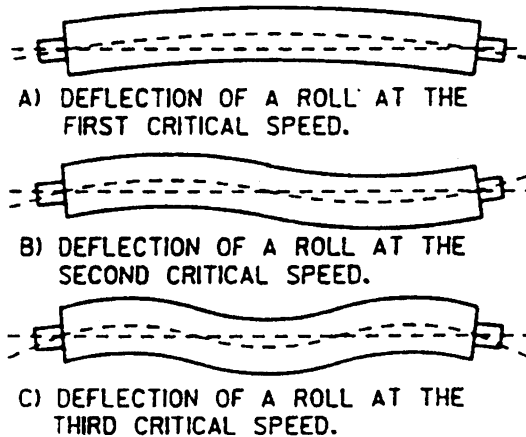


FIGURE 3
MODE SHAPES OF FIRST, SECOND AND THIRD CRITICAL SPEEDS
FOR A SIMPLY SUPPORTED ROTOR

4.3 IMPACT/IMPULSE NATURAL FREQUENCY TESTING

Of all the testing techniques to measure natural frequency, impact testing is the most popular even though it might not provide the ultimate precision of the other methods discussed. For one thing, it can even be performed with fair accuracy using a single channel spectrum analyzer or data collector and a rubber mallet (or even a 4" X 4" piece of lumber). A "dead blow" or rubber mallet is generally used to repetitively tap the structure or machine. The mallet or hammer has to impact with enough force to at least produce some amplitude response so the effect can be seen. The amount of amplitude is not really significant in this test (unless one "overdrives" the input of the analyzer causing it to truncate the time waveform, thereby introducing a series of "false frequencies" and additional noise). The harder one strikes, the higher the amplitude. What really is significant is measuring the frequency or frequencies which will appear.

Use of a rubber mallet will excite all natural frequencies in the range from about 0 - 30,000 CPM (500 Hz) or higher, depending on the hardness of the rubber. Commercially available units called "modally-tuned hammers" can equally excite frequencies out to about 120,000 CPM (2000 Hz). That is, when one of these elastomer-tipped hammers impacts the structure, it will impact roughly equal energy across a broad range of frequencies up to a certain "cut-off frequency" where the response will drop off as shown in Figure 4. If any natural frequencies are present within this "flat response region" (Figure 4), they will be excited by the impact and will respond. Importantly, the tip material and hardness will limit the highest natural frequency which can be excited as suggested by Figure 4 (remember, do not use aluminum or steel-tipped hammers for they will corrupt the frequency response due to the "double hits" they almost surely will impart) (Ref. 4).

If a dead blow hammer or rubber mallet is not available, a four-by-four timber about five feet long is adequate for simple impact testing with a single channel analyzer. Sledge hammers with steel heads, unless covered with an elastomer, are not adequate since "double hits" or even "triple hits" again introduce noise into the frequency response. These hammers bounce off the structure and do not usually impart enough energy into the system (Ref. 5).

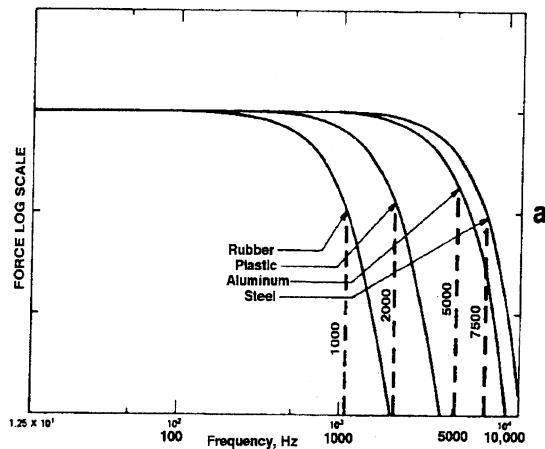


FIGURE 4
FORCE SPECTRA PRODUCED WITH VARIOUS HAMMER TIP MATERIALS

For this type of testing, older technology instruments can be used with some success. Tunable-filter analyzers (i.e., IRD 350) with a frequency meter could be used. The analyzer is placed in the "filter-out" mode during the impact. If the natural frequency is distinct, it will appear on the meter². Then, as the analyzer is tuned through various frequency ranges, the natural or resonant frequencies can be located as the machine or structure is repeatedly tapped. Similarly, a swept-filter analyzer (i.e., IRD 880) could also be used for structural impact testing. The structure is *struck continuously* as the analyzer sweeps automatically through a selected frequency range². Here, it will be important to attempt to impact the structure with about the same force, to allow it

FIGURE 5
IMPACT TESTS ON TWO MACHINE TOOL SPINDLES USING A SWEPT-FILTER ANALYZER

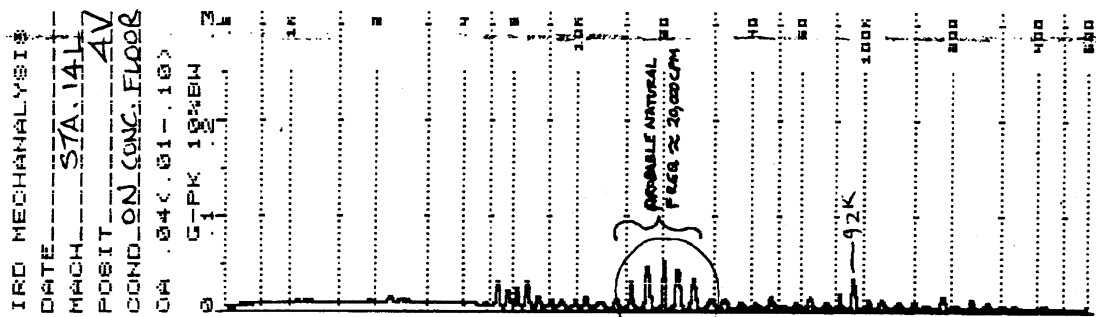


FIGURE 5A. THE NATURAL FREQUENCY IS ABOUT 20,000 CPM.

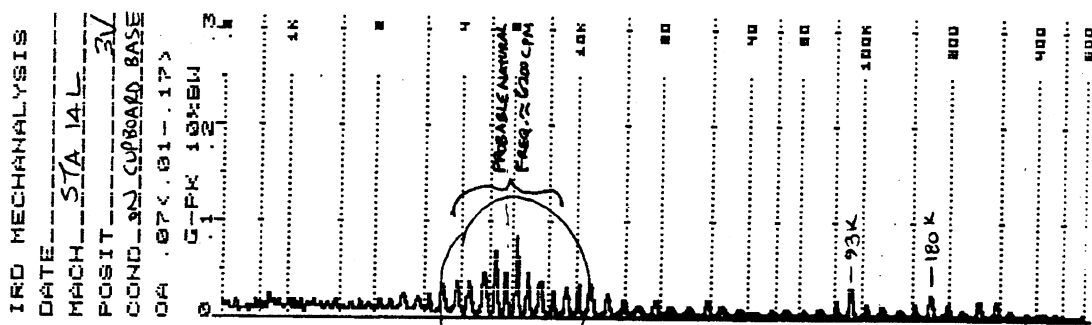


FIGURE 5B. THE NATURAL FREQUENCY IS ABOUT 6200 CPM.

to "ring down", time for the analyzer to slowly sweep through the frequency range, and to ensure once again one does not overdrive the analyzer electronics for valid results. In Figure 5 are amplitude versus frequency strip-charts from an IRD 880 swept-filter analyzer. A "bump" (impact) test was done on two precision machine-tool box spindles to determine their natural frequencies. A dead blow hammer was used to strike the housing in the vertical direction near the bearings. Note that in Figure 5A the natural frequency was around 20,000 CPM; and in Figure 5B the natural frequency was around 6200 CPM for that particular box spindle assembly.

However, note that the measurement conditions were different for each box spindle assembly. The one in Figure 5A was struck while it was resting on a concrete floor whereas the spindle in Figure 5B was struck while it was resting on a steel cabinet base. Although both box spindles were struck in the vertical direction, the first one was struck at the bearing nearest the tooling and the second one was struck at the bearing nearest the drive end. The point is, in order to have comparative natural frequency tests, the conditions have to be identical. The box spindles have to be identical and they have to have the same base mounting method. They also need to be impacted in the same direction with the same mallet and on the same location. If not, all the natural frequencies will be different in an impact test. That is, the natural frequencies in the horizontal and axial directions will both be different than the natural frequencies in the vertical direction. In addition, the natural frequencies which will be measured will be the "system" natural frequencies, not of any particular machine element alone. The natural frequencies measured at the drive end can be different than the natural frequencies measured at the tool end, especially if one box spindle is longer than the other.

Instead of using dead blow hammers or rubber mallets, a more sophisticated approach would be to use a "modally-tuned" hammer. Various models of modal hammers are available like the one shown in Figure 6. These can range in size from one with a .005 lb steel head for testing very light structures such as printed circuit boards; to those with 1.0 lb steel heads for general rotating machinery and medium to heavy structures; and on up to "sledge hammers" with 12 lb steel heads for exciting very heavy structures like buildings, ships and foundations. Note that a force transducer is placed just behind the hammer tip with which the impact force can be measured and the signal transmitted over to an analyzer through an electrical connector normally located at the end of the hammer's handle. Also note that a number of tips are supplied with each hammer. These tips vary in hardness from "soft" to "hard" to excite the frequency range of interest (Ref. 6).

The modal hammer can be used in conjunction with either a single-channel or multi-channel data collector or spectrum analyzer. In this case, it is very important to allow the structure to "Ring Down" before hitting it again. With this analyzer, the impact tests can be performed in either the time or frequency domains. During the impact tests a uniform (no window) should be used on the analyzer. Some analyzers have special windows for impact tests (such as a force/exponential window) (Ref. 5).

Triggering can be "free run" or from a hammer source. Triggered data capture simply means that the analyzer only collects data when the amplitude level exceeds a specified trigger level. In addition, many analyzers have the capability to automatically reject data if the electronics are over-driven by too great an impact (this prevents great corruption of the data).

Figure 7 shows the results of an averaged impact test. In this case the analyst was measuring the principal horizontal natural frequency of a 100 HP DC motor driving a friction roll assembly through a gearbox. Four impacts with a 4" X 4" timber were averaged to reduce random signals (Ref. 3). Note there were three responses at 38.9 Hz, 42.0 Hz and 90.4 Hz (corresponding to 2334, 2520 and 5424 CPM, respectively) which likely corresponded to the first, second and third horizontal natural frequencies. Note that the raised "peak" on the lower, left-hand side at about 8 Hz (480 CPM) was likely only integration noise just above the lower cut-off frequency of the instrument, although it is possible this might have been close to what is known as a "rigid body mode" of the structure (which is discussed in Section 4.424).

FIGURE 6
AN EXAMPLE OF A MODALLY-TUNED HAMMER (Ref. 6)



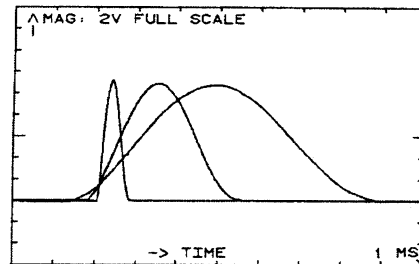
**impulse hammer
kits, series
291B**
for
**TESTING
STRUCTURAL
BEHAVIOR**

- Measures stimulus (force) and response (motion)
- Excites vibrations and resonances, quickly and conveniently
- Adapts FFT analyzers and computers for modal analysis
- Helps search resonances, detect faults, check quality, save material, reduce weight, improve life, modify behavior and create or verify models
- Includes quartz sensors with built-in amplifiers designed specifically for impulse testing

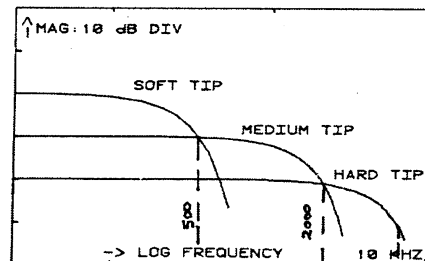
Now proven in thousands of applications, the PCB impulse hammers simplify structural testing. They function to transfer force into an electrical signal. As illustrated, the frequency spectrum of the time-varying impulse signal is nearly flat over a wide frequency range, similar to that of a random excitation. This transformation indicates that a simple impulse excites all resonances within a frequency band. These kits include quartz accelerometers. They function to transfer motion caused by force into an electrical signal proportional to the acceleration aspect of motion. They also serve as a reference for hammer calibration.

With the variety of tips and extenders supplied, you can tailor the hammer structure to fit your application. The different hammer structures available enable you to test a variety of structures - from small objects such as PC boards and sheet metal parts to objects as large as ships and buildings. The hammer structure governs its behavior. The length, material, cushioning, mass and impact velocity shape the impulse waveform and determine its frequency content. Some of these university-aided-design, modally tuned PCB hammers employ high modulus graphite fiber handles and special grips to improve behavior. In general, electrical connectors are located at the end of the handle.

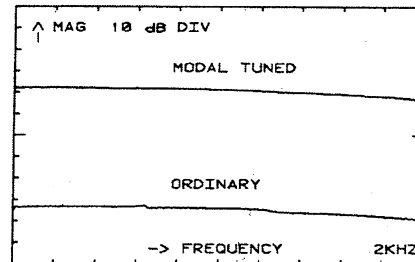
Modal analysis involves impacting the test object with the force instrumented hammer at many points and measuring the motion at one critical point (or vice versa). These tests supply the stimulus and response information to compute classical transfer characteristics like inertance, compliance, impedance or mobility. The complete task of determining vibration mode shapes requires a sophisticated dual-channel FFT analyzer. Simple analyzers and scopes provide partial answers. A variety of other PCB sensors (force, motion and driving point) adapt analyzers for testing with random sinusoidal or multiple input excitation.



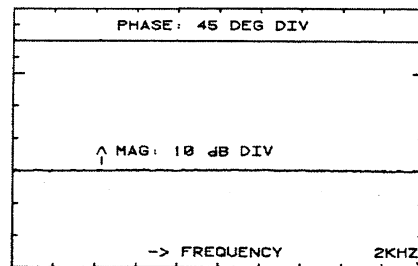
History of time-varying signals from different hammer structures (different extenders and tips)



Spectrum of impulse signals from different structures (hard, medium and soft impact tips)



Comparison of hammer structures



Typical ratio calibration of hammer

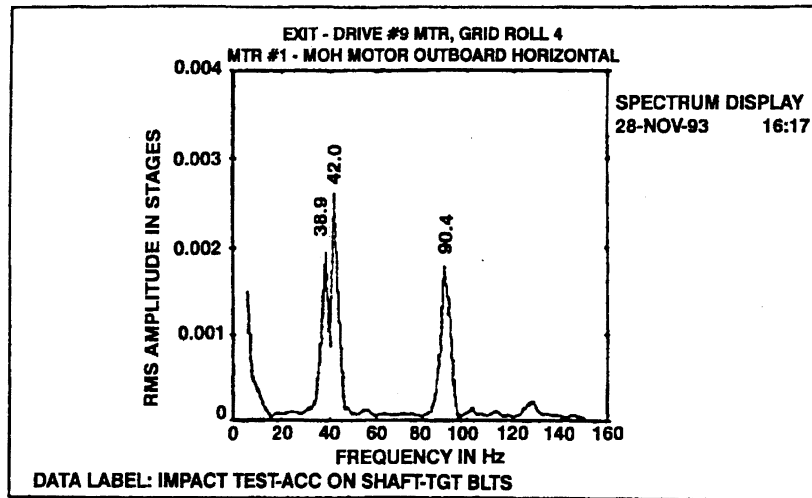


FIGURE 7
RESULTS OF AN IMPACT TEST ON A 100 HP DC MOTOR
EMPLOYING AVERAGING (Ref. 7)

Averaging several impacts improves data quality, provided that a consistent impact force over the same time duration is maintained. *The single-channel analyzer cannot provide a precise estimate of the vibration response per unit load or the relative phase between response and load.* By directly measuring the force spectrum, an *estimate of average* impact force level is possible. However, without the phase between the force and response, *only crude estimates of mode shape can be made.*

A short impulse excites modes over a broad frequency range. The force pulse produced from the impact is of short duration. If it is shorter than the length of the time record, the force spectrum is quite flat at lower frequencies, as shown in Figure 8.

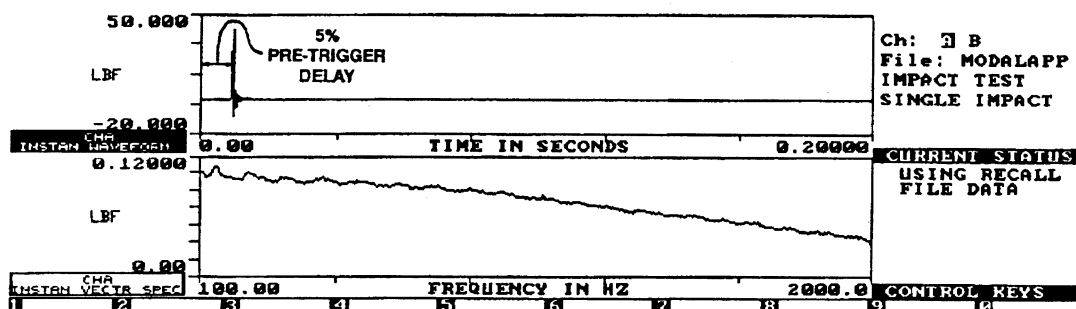


FIGURE 8
A FORCE PULSE CAUSED BY SHARP IMPACT SEEN IN TIME WAVEFORM
WILL PRODUCE A FLAT FORCE SPECTRUM AT LOW FREQUENCIES (Ref. 8)

On the other hand, Figure 9 shows the vibration response spectrum from the impact. Natural frequencies and damping can be *estimated* from the position and width of the response peaks in the spectral display. *The important thing to observe here is one needs to look at the time domain to ensure the response has returned to (or near) zero in order to ensure he has good spectral response data by minimizing leakage effects.* If it has not he may have to allow more time (t_{MAX}) for the structure to do so, or may have to employ one of the special exponential windows now available on some analyzers on the force response to force it to zero.

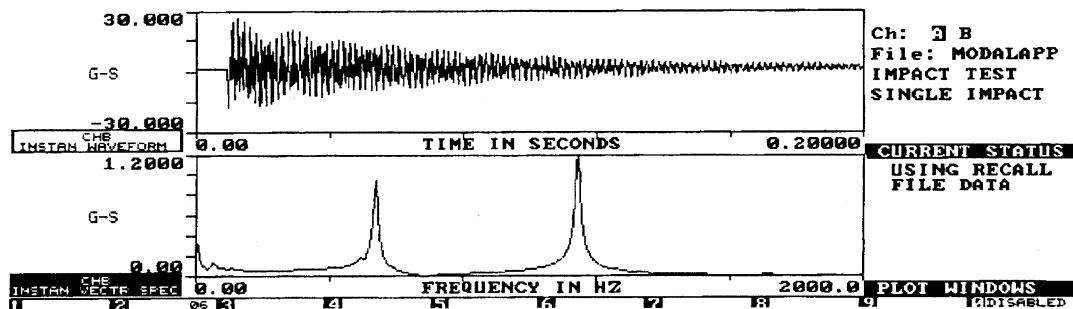


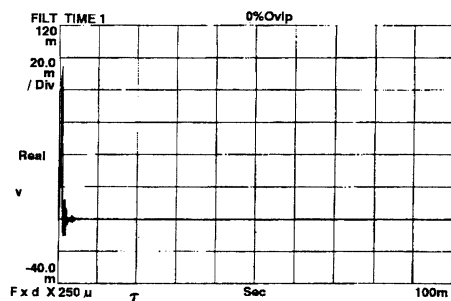
FIGURE 9
VIBRATION RESPONSE TO IMPACT SHOWING NATURAL FREQUENCIES
IN THE SPECTRUM AND THE "RING-DOWN" IN THE TIME WAVEFORM
(Ref. 8)

While the force signal from an impact is almost always of short duration, there may be a problem if there is background noise from surrounding machines or if there is electronic noise in the analyzer. Each will contribute to the total signal, apart from the impact, over the remainder of the time record. The signal-to-noise ratio will therefore be reduced, reducing the clarity of the test.

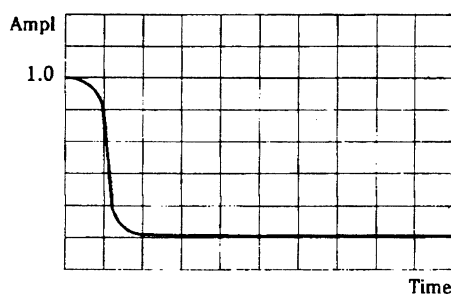
Impact testing produces transient signals. If a Hanning window is used, it will distort the signal since it automatically "forces" both the beginning and ending of a time block to near zero. For correct results, a transient signal must start and finish within the time block of an FFT analysis. As was mentioned previously, a Uniform window (no window) is the correct way to treat the signal before computing the spectrum. Smoothing windows like the Hanning or Flat Top windows are intended for continuous signals.

To ensure the impact test excites as many natural frequencies as possible, acceleration should be used rather than velocity or displacement. Higher and higher natural frequencies normally will have lower response. Acceleration data will help "elevate" them in the instrument display.

To minimize the effect of instrument or background noise, a *force window* can be applied. It zeroes data values before and after the impact and applies a unity (1.0) weighting factor during the impact. An example force window is shown in Figure 10 which sharply filters the response just after impact. Sometimes, similar noise effects can be reduced in response analysis by zeroing the data before the trigger (Ref. 8).



Force Pulse

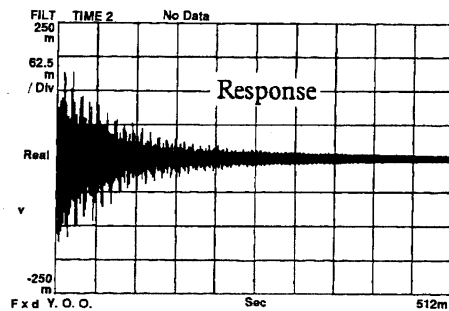


Force Window

FIGURE 10
FORCE PULSE WITH WINDOW APPLIED (Ref. 9)

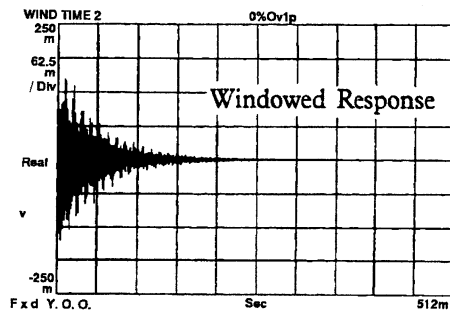
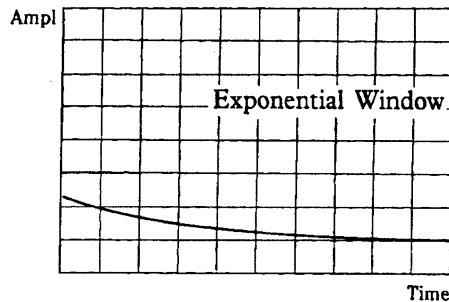
Apart from these effects, the vibration response signal may not always die away before the end of the time record, especially for lightly damped structures. The resulting truncation of the signal can cause significant leakage in the spectrum. This can lead to poor estimates of natural frequencies, damping, and response. To overcome this problem, an exponential window may be used, which applies an exponentially decreasing weighting to the response signal. An example exponential response window is shown in Figure 11. With sufficient weighting, the response signal can be reduced to an insignificant level by the end of the time record. In lightly damped systems, where the natural frequencies not affected very much by damping, exponential weighting can still be achieved effectively without affecting the natural frequencies significantly. Even for strongly damped systems, exponential weighting is sometimes useful to minimize the effects of background noise in long time records.

The setup of a Force/Exponential window combination for impact testing must be linked to the trigger delay setting, or the signals will be distorted by the windows. Pretriggering must be used, with a delay that puts the whole of the impulse waveform within the time record, equal to a small fraction of the length of the time record. A delay of 5 to 10% of the record length is sufficient. The



Response Continues
To "Ring Down"
Beyond Time Block
(Not Desirable)

Exponential Decaying Response



Response Is Artificially
Forced "Ring Down"
By The Exponential
Window

FIGURE 11
APPLICATION OF AN EXPONENTIAL WINDOW TO THE RESPONSE
WAVEFORM TO FORCE IT TO GO TO ZERO BEFORE THE END
OF THE TIME RECORD (Ref. 9)

analyzer is started, causing data to be acquired continuously, until the impact force signal exceeds the trigger level. The analyzer does not stop at this point, but continues to acquire data until a complete time record block is filled. The trigger point is positioned inside the time block at the specified delay time from the beginning. Typically, the FFT is then computed, the spectrum averaged and the cycle is repeated, waiting for a new impulse to exceed the trigger level, before another spectrum is computed. The spectrum shown in Figure 9 was set up to capture the force pulse with a pre-trigger delay of 5% of the record length. The force window must begin shortly before the trigger and end shortly after the impulse. A quick test will confirm.

The exponential window can start at the same point as the force window. The exponential decay rate must be sufficient to suppress the signal by the end of the record. As a rule of thumb, the decay constant should be at least 1/4 of the length of the time record, with a short trigger delay. This reduces the signal to less than 2% of its unweighted value. Provided that there are no resonances that lie within the first 10 lines of the spectrum, there are no limitations on the frequency range or the resolution if exponential weighting is used (Ref. 4).

It was mentioned previously that averaging several impacts improves data quality provided that a consistent impact force over the same time duration is maintained. If one "hit" with a hammer is much "harder" than the rest, it is called "over-hitting" and will cause an overloaded force or response signal. If one "hit" is much "softer" than the rest, it is called "under-hitting" and will cause an under-ranged force or response signal. Another common problem is "double-hitting". Typically with lightly damped structures, the structure actually rebounds before the hammer is withdrawn. This produces an uneven force level with frequency. Don't forget to allow structure to "Ring Down" before hitting it again.

All of these effects tend to corrupt the results and should be eliminated from the average. Some analyzers allow each block of data to be reviewed before being accepted, while others automatically suppress over-ranged or under-ranged data. Usually, double-hitting can be detected in the force input signal and bad data is rejected, but this is inconvenient.

In conclusion, using a swept-filter analyzer with a dead blow hammer or a single-channel analyzer with a modally-tuned hammer does not give precise natural frequencies. They provide reasonable, useful estimates of the natural frequencies. If more efficient, more precise, and more informative data is required, then a 2-channel analyzer should be used to simultaneously provide both phase and frequency information using cross-channel measurements and a coherence function.

4.4 RUNUP AND COASTDOWN NATURAL FREQUENCY TESTS

One of the strong points about runup and coastdown tests is that they actually evaluate the machine while it is under actual running dynamic conditions. Then, if there are any dynamic changes to natural frequencies due to gyroscopic or fluid-film pressure/stiffness changes, these will be taken into account, particularly if the machine is allowed to come to full operating temperature. Gyroscopic effects can sometimes be considerable on overhung rotors, particularly if the overhang is significant. Such gyroscopic effects can cause natural frequencies to increase due to dynamic "stiffening" of the rotor.

Runup and coastdown tests can be performed with a wide array of instruments and employ a range of techniques. In essence, they can consist of nothing other than watching the overall vibration with a hand-held meter, observing how it changes as the machine coasts down from full speed. This is somewhat meaningful, but has limited useful applications. On the other hand, techniques such as Bode' plots and Polar diagrams can be applied to natural frequency testing to track and display any changes as the machine either accelerates from zero to full speed, or coasts down to a stop. Following below will be a description of how to acquire and interpret Bode' plots and Polar plots, both with and without using a tracking filter:

4.41 BODE' PLOTS

The Bode' Plot actually consists of two separate Cartesian plots - (1) Amplitude versus RPM; and (2) Phase Lag versus RPM. When a machine passes through a natural frequency during a coastdown, the Bode' Plot will indicate two changes: (1) the amplitude plot will show a peak, and (2) the phase plot will show a 180° phase shift (with a phase shift of 90° right at the natural frequency).

4.411 A Tracking Filter is Needed for Bode' Plots

A tracking filter utilizes "phase-locking" techniques to be able to record amplitude versus RPM and phase versus RPM information. If a 2-channel analyzer is used, the amplitude versus RPM plot is done on one channel while the phase versus RPM plot is done on the other channel so that each peak is displayed at an RPM which corresponds to the machine RPM at that exact moment, a reference pickup is used. The reference pickup is placed at the rotating shaft to provide a voltage pulse for each revolution of the shaft. Photocells and electromagnetic pickups are normally used to provide the reference pulse. An existing key or keyway on the shaft is an excellent target for them. If none are available, a piece of reflective tape can be used with a photocell.

The reference signal automatically tunes the tracking filter to the RPM of the shaft. If the shaft RPM changes, the frequency of the reference signal is also changed accordingly. In this way, the analyzer's filter is locked or synchronized to the shaft RPM. Also, the reference signal provides a fixed reference for comparison with the signal from the accelerometer. The result is a DC voltage proportional to the relative phase between the two signals. This DC voltage indicates a relative phase measurement (0° to 360°) on the phase versus RPM plot (Ref. 10).

4.412 Explanation of a Bode' Plot

A sample Bode' Plot is shown in Figure 12. It is apparent that the machine in question has two significant natural frequencies, one at approximately 2450 RPM and another at 4850 RPM. Of course, if there is an exciting force which corresponds to either of these frequencies, this will cause the machine to operate in resonance. These forces might be due to normal unbalance, misalignment, aerodynamic or hydraulic forces, torque pulses, reciprocating

forces, looseness, oil whirl, etc. Then, an undesirable and probably destructive vibration can likely result. For this reason, it is helpful to know:

- a) What exciting frequencies of vibration are inherent to the machine at its normal operating speed(s). These can be identified through normal amplitude-versus-frequency analysis techniques already discussed.
- b) What are the natural or resonant frequencies of the machine or structure. As illustrated in Figure 12, a plot of amplitude and phase versus machine RPM will reveal these (Bode' plot).

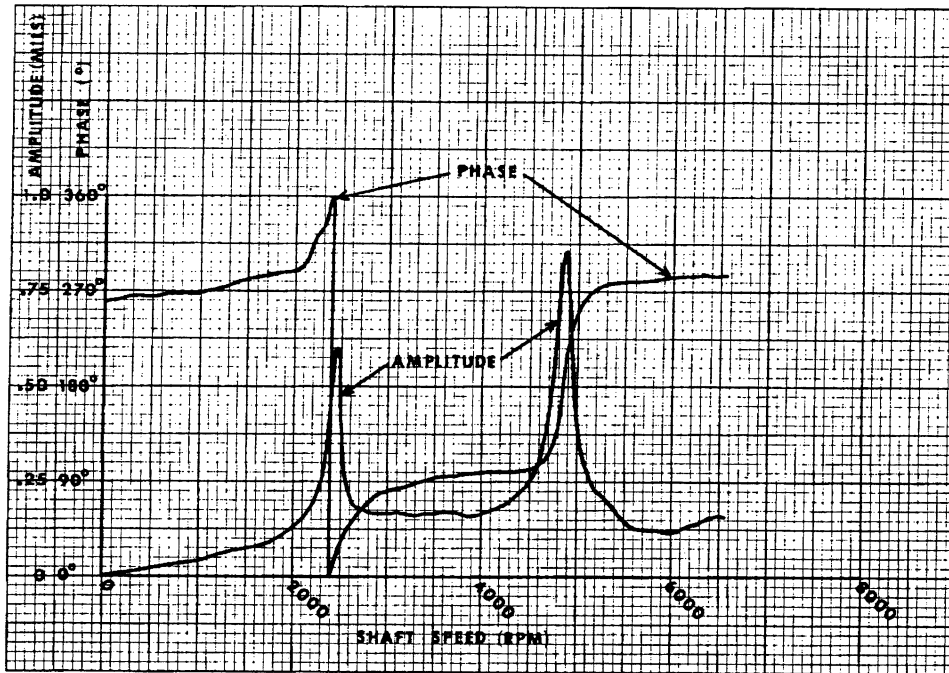


FIGURE 12
EXAMPLE BODE' PLOT USED TO CONFIRM THE LOCATION
OF NATURAL FREQUENCIES (Ref. 10)

4.413 Interpreting Unusual Bode' Plots

As mentioned previously, when a machine passes through a natural frequency during coastdown or start-up, the usual indications will be a peak in amplitude at the resonance speed accompanied by the characteristic phase shift of approximately 180° (see Figure 12).

Occasionally, however, the recorded data may reveal some unusual conditions of system response as illustrated by the following examples:

EXAMPLE #1:

Referring to the data in Figure 13, two distinct amplitude peaks are noted at about 500 RPM and 1200 RPM suggesting two resonant speeds. However, the recorded phase data shows the characteristic 180° phase shift for only the lower speed (500 RPM) amplitude peak. There is virtually no phase shift associated with the higher 1200 RPM peak. Based on the recorded

data, it can be concluded that the lower 500 RPM peak is, in fact, a true resonance. However, the 1200 RPM amplitude peak is not resonance due to the lack of a corresponding phase shift. The question now is; "What would cause a peak in 1X RPM vibration which is not due to resonance?"

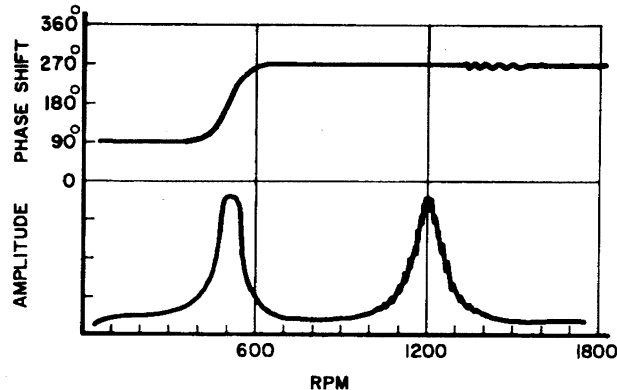


FIGURE 13 (Ref. 10)

One possible cause of the 1200 RPM amplitude peak in Figure 13 might be the presence of significant background vibration at the indicated frequency. For example, suppose a plot were being taken during the coastdown of an 1800 RPM motor, and at the same time, a 1200 RPM fan operating nearby was contributing substantial background vibration at 1200 CPM. Obviously, as the 1800 RPM motor coasts down, its speed must eventually coast through 1200 RPM. When this happens, the motor vibration will momentarily add to the background vibration from the fan to produce an amplitude peak at 1200 RPM.

EXAMPLE #2:

Referring to the amplitude and phase plots in Figure 14, an amplitude peak is noted at a speed of approximately 600 RPM.

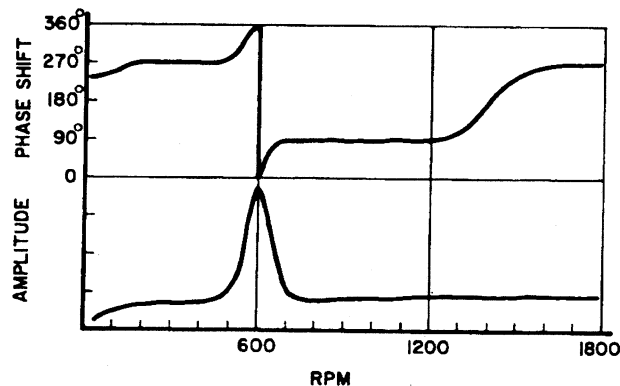


FIGURE 14 (Ref. 10)

A 180° phase shift corresponding to the amplitude peak confirms this as being a resonance. In addition, there is also a 180° phase shift at approximately 1400 RPM - but there is no amplitude peak at this speed to suggest resonance. The 180° phase shift, however, verifies that there is, in fact, a resonance at 1400 RPM. The question here is; "How can there be a resonance without a peak in amplitude?" The absence of the amplitude peak at resonance might be the result of one of the following:

- a) If the exciting force corresponding to the resonant frequency is very low, or if the system is *heavily damped*, little amplification in amplitude may occur at resonance. (See Figure 15 for the effect of damping on system response).

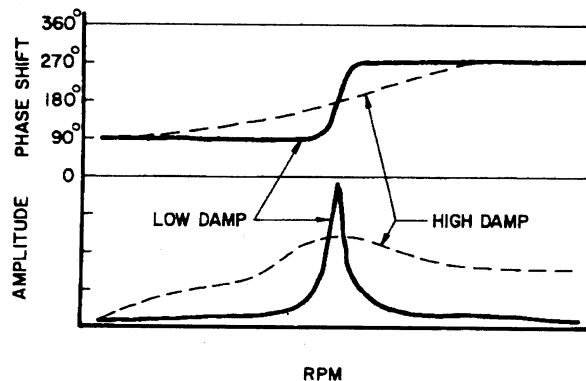


FIGURE 15
HIGHLY DAMPED VERSUS A LOWLY DAMPED RESONANT RESPONSE
(Ref. 10)

- b) *If the transducer is located at a "nodal point" of the resonant system, little or no noticeable increase in amplitude may be noted at the resonance frequency. When a rotor shaft or structure is excited to vibrate at resonance, it will likely assume one of the vibratory modes illustrated in Figure 16. Each mode has one or more nodal points which are points of minimal amplitude. Of course, if the transducer is located at a nodal point, there may be little noticeable amplification of the vibration amplitude when passing through this resonance. However, the phase plot will reveal the characteristic 180° shift. To prevent this from causing one to miss potential natural frequencies, the structure should be impacted in several different locations as shown in Figure 17.*

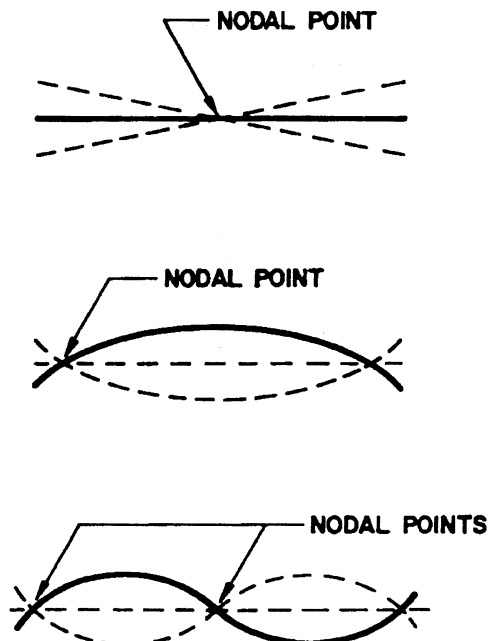
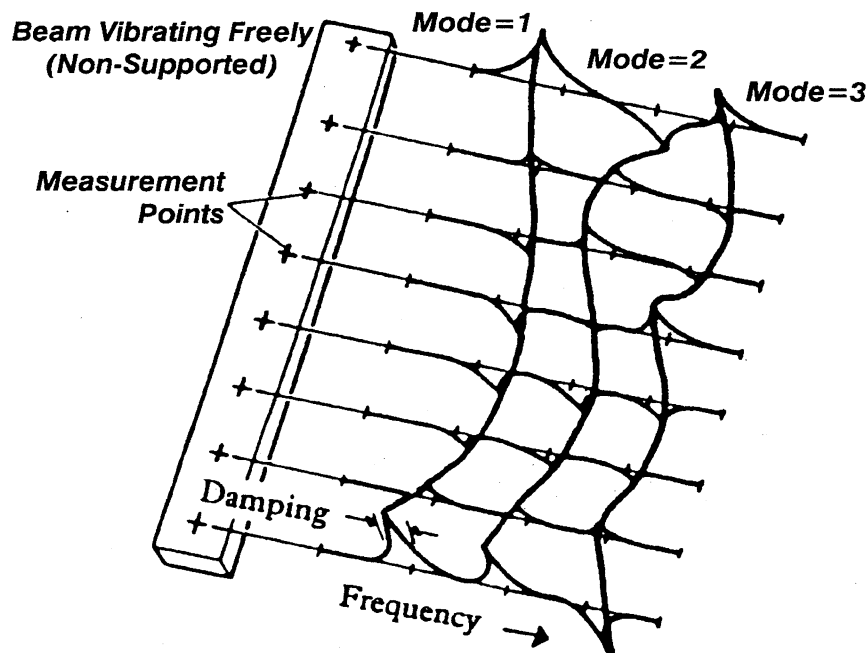


FIGURE 16
NODAL POINTS (Ref. 10)



Damping, Frequency - Same at Each Measurement Point.

Mode Shape - Obtained at Same Frequency From all Measurement Points.

FIGURE 17
EXCITATION OF SEVERAL MODES AT VARIOUS POINTS OF A STRUCTURE (Ref. 4)

EXAMPLE #3:

Referring to the amplitude and phase-versus-RPM data in Figure 18, it is noted that an amplitude peak occurred at approximately 900 RPM. However, checking the phase information it is noted that the amplitude peak is accompanied by a phase shift of approximately 360° instead of the usual 180° . In this case, the phase data suggests that there are actually two systems in resonance at or near the same frequency, and each of the two resonances is contributing a 180° phase shift for a total phase shift of 360° (these are called closely coupled modes). In this case identifying and correcting only one of the resonant systems may not totally solve the problem. One must use either more lines of resolution and/or a smaller frequency span to separate these 2 natural frequencies.

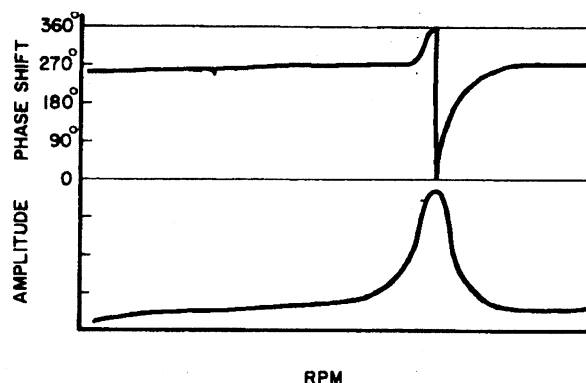


FIGURE 18 (Ref. 10)

EXAMPLE #4:

Referring to the sample data in Figure 19, the somewhat unusual portion of the data which deserves explanation is the "dip" in vibration amplitude which occurs at approximately 1000 RPM; along with the accompanying 360° shift in phase. As in EXAMPLE #3, the 360° phase shift at 1000 RPM in Figure 19 suggests two spring-mass systems in resonance at or very near the same frequency. The dip or reduction in vibration amplitude at this frequency, sometimes referred to as a "anti-node" can be explained in Figure 20.

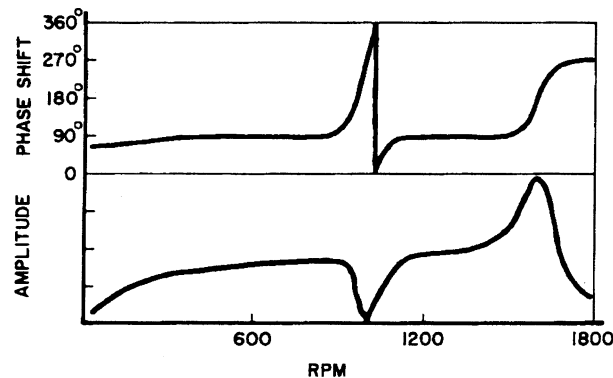


FIGURE 19 (Ref. 10)

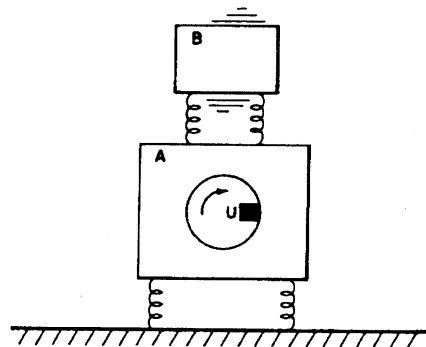


FIGURE 20
DYNAMIC VIBRATION ABSORBER (Ref. 10)

First, assume that the initial exciting force is the unbalance, "U". If spring-mass system "A" is resonant with the unbalance frequency, then we know from our previous discussions that the actual vibration of mass "A" lags the unbalance force by 90°. Now, if spring-mass system "B" is resonant at the same frequency, then its exciting force will be derived from the motion of mass "A". Therefore, the actual vibration of mass "B" must lag the motion of "A" by 90°. As a result, the actual vibration of mass "B" will lag the initial unbalance force by a total of 180° ($90^\circ + 90^\circ = 180^\circ$). This produces two opposite forces acting on mass "A", one being the initial unbalance force and the second being the force of vibration of mass "B". Since the two forces acting on mass "A" are opposite, mass "A" will exhibit a minimal vibration amplitude as illustrated by the dip in Figure 19. Mass "B", on the other hand, may reveal an extremely high amplitude of vibration. For this reason, when the data suggests a possible anti-node, it would be advisable to identify the second system in resonance and plot its response as well, to insure satisfactory performance. Figures 19 and 20 illustrate how and why a dynamic vibration absorber is often very effective.

EXAMPLE #5:

Plots similar to that in Figure 21 are sometimes obtained when non-contact pickups are used to identify rotor critical speeds. The plot does display a peak amplitude and the normal 180° phase shift. However, it should be further noted that the amplitude plot includes a "dip" at approximately 5000 RPM, and, in this case, just slightly below the shaft critical speed. The amplitude dip in Figure 21 does not have a substantial corresponding phase shift and, thus, is not likely the result of an anti-node, plotted in Figure 19. The cause of the indicated amplitude reduction (dip) in the plot, Figure 21 can usually be traced to excessive electrical and/or mechanical run-out of the shaft at the non-contact pickup target area. The non-contact or proximity pickup cannot distinguish between actual shaft vibration and any run-out or eccentricity of the shaft journal. As a result, the non-contact pickup provides a signal proportional to the vector sum of run-out and actual shaft vibration.

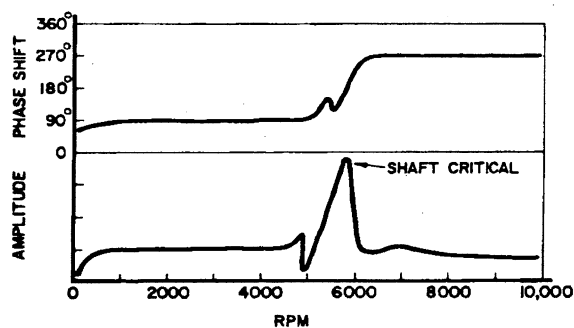


FIGURE 21 (Ref. 10)

Operating well below its critical speed, a shaft is considered to be rigid and, thus shaft vibration will essentially be in-phase with unbalance. However, as the RPM of the shaft approaches its critical speed, shaft deflection will progressively increase. In addition, the deflection in the shaft (actual shaft vibration) begins to lag the heavy spot of unbalance. In fact, operating at critical speed, shaft deflection lags the unbalance heavy spot by 90° . and at operating speeds above shaft critical, deflection will lag the unbalance by 180° . The important point to note here is that the amplitude and phase of shaft vibration is undergoing a change as the shaft passes through critical speed. As a result, the vector sum of shaft vibration and run-out is also undergoing a change. The dip or amplitude reduction plotted in Figure 21 results when the shaft vibration and run-out vectors become momentarily out-of-phase and, thus, tend to cancel one another. Of course, instead of canceling one another, it may occur that the run-out and shaft vibration vectors will momentarily add to reveal a plot such as that in Figure 22. In either case, it is important to note that when using non-contact pickups, the presence of excessive run-out may give a very distorted picture of rotor response. The recorded data may suggest a condition which is far better or far worse than the true response of the rotor. Therefore, it is advisable to carefully measure run-out amplitude before response measurements are taken. Of course, excessive run-out should be physically or electronically eliminated so that plots of true rotor response can be obtained.

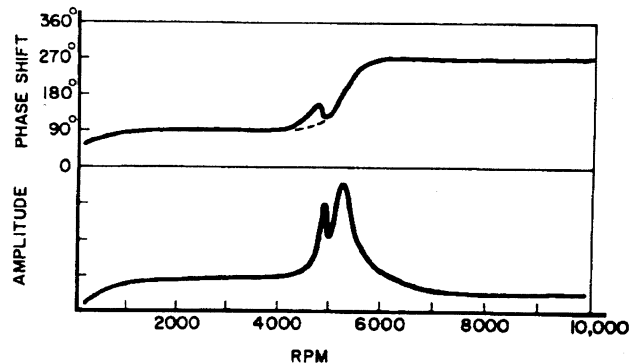


FIGURE 22 (Ref. 10)

EXAMPLE #6:

The recorded plot in Figure 23 illustrates a situation where the recorded data can be somewhat misleading. The suspicious data in the plot, Figure 23, are the somewhat unusual phase shifts that occur at 6000 and 7000 RPM. While these apparent phase shifts may suggest resonant frequencies which were not indicated on the amplitude plot, it is quite possible that these are not phase shifts at all. From our previous discussion you will recall that two signals are required to obtain a DC voltage proportional to phase for plotting purposes. These signals are 1) a voltage pulse from a reference pickup at the shaft of the machine and, 2) a vibration signal from a vibration pickup. If either one of these signals is lost or lacks sufficient amplitude, the phase voltage will drop to zero. Referring to the amplitude plot in Figure 23, you will note that the vibration amplitude is at a relatively low value between 6000 and 7000 RPM. Note also that the phase plot drops to 0° between 6000 and 7000 RPM. Therefore, it is more likely that the apparent phase shifts are nothing more than the absence of a sufficient vibration signal to provide a true phase reading. Below 6000 and above 7000 RPM the vibration amplitude has increased sufficiently to provide phase indications.

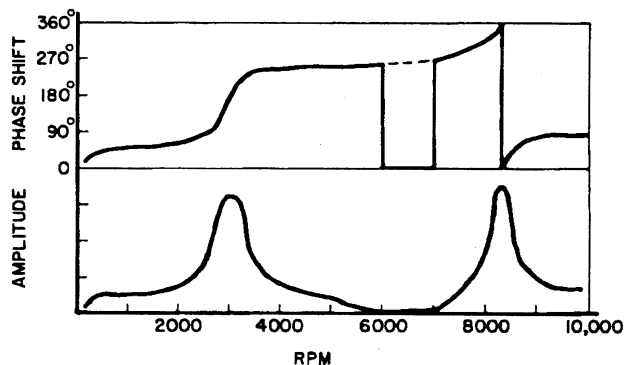


FIGURE 23 (Ref. 10)

Bode' Plots are not without limitations. While they readily reveal natural frequencies of the machine or structure or the entire spring-mass system, Bode' plots alone may not reveal the response of the system to other vibration frequencies which can be affecting the overall performance of the machine. The analyst needs to be aware of this. Remember, they are taken from filtered data which shows the response of this system as the speed (or frequency) is brought down or up through these frequency regions.

4.42 POLAR PLOTS

In the previous section, Bode' Plots were used to identify natural frequencies or resonances. Recall that Bode' Plots were amplitude-versus-RPM and phase-versus-RPM plots. These were recorded on a Cartesian (rectangular) coordinate graph.

Another technique for presenting this information is called the "polar" plot. Instead of using a rectangular graph, polar plots are displayed on a polar coordinate graph. The polar plot is a presentation of amplitude and phase only, unlike Bode' plots.

4.421 Setting Up for Polar Plots (Ref. 10)

As in the case of Bode' plots, the instrument used needs to be equipped with a tracking filter. It also needs polar amplitude and phase output signals to drive a standard X-Y recorder.

The pen on the X-Y recorder needs to start at the center point of the polar graph. To do this the X-Y recorder is adjusted so that a zero-amplitude signal will position the pen at the center of the graph. The FULL SCALE amplitude control on the recorder is adjusted so that as the vibration amplitude increases it will cause the pen to move radially outward from the center. The amount that the pen moves away from the center is proportional to the amount the vibration increases. The phase of the vibration will determine the angular direction in which the pen will move as the vibration increases.

4.422 Advantages of Polar Plots Over Bode' Plots (Ref. 10)

1. Provides an immediate indication of the unbalance vector without the need to compare separate amplitude and phase plots.
2. Eliminates confusing phase discontinuities that appear on Bode' plots as the phase changes from 0° to 360° (which occurred at the 2400 CPM natural frequency on Figure 12).
3. Eliminates phase discontinuities that appear on Bode' plots when the vibration amplitude has reduced to a low level where there is not sufficient vibration signal to maintain a phase indication. See Figure 23.
4. Where only a single-pen recorder is available, the Polar plot allows both amplitude and phase data to be obtained during a single run-up or coastdown of the machine, since a Polar plot only requires a single pen anyway.
5. Where non-contact pickups are being used, the vector resulting from combined mechanical and electrical run-out can be eliminated by recalibrating the recorder amplitude ZERO control to the origin of the polar graph while the run-out vector is being displayed with the machine on slow-roll. Figure 24 shows comparative Polar plots obtained with and without recalibrating for shaft run-out. As illustrated, run-out can result in substantial unbalance vector at any one operating speed.

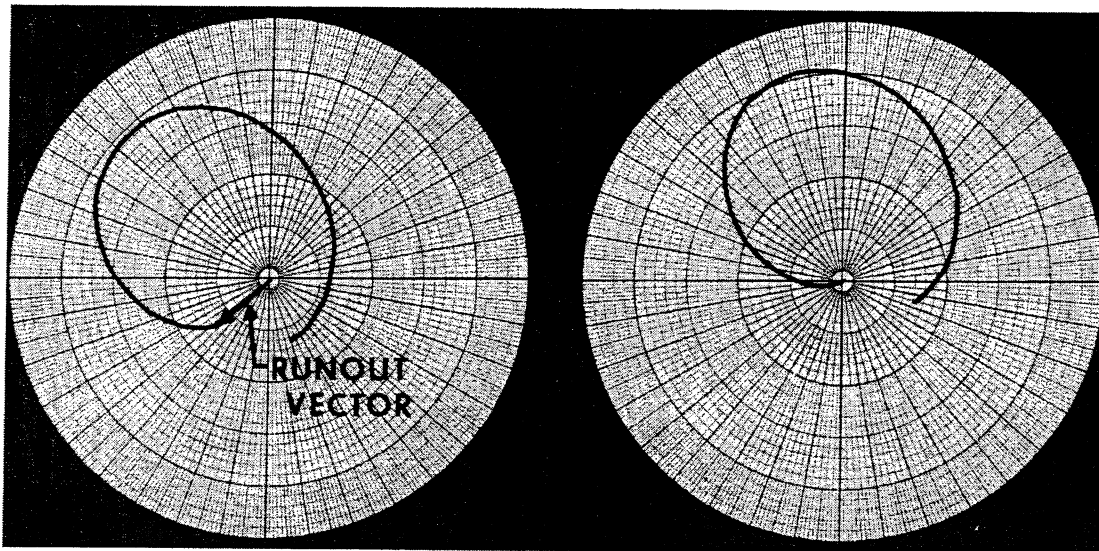


FIGURE 24
POLAR PLOTS WITH AND WITHOUT RUNOUT COMPENSATED (Ref. 10)

4.423 Comparison of Bode' and Polar Plots for Natural Frequency Testing

The plots in Figure 25 illustrate how a polar plot compares with a familiar Bode' plot. Note that in the Bode' plot, there are two distinct peaks in the lower, amplitude-versus-RPM plot. One is at approximately 2400 RPM and the other is at approximately 4900 RPM. Are these peaks resonant?

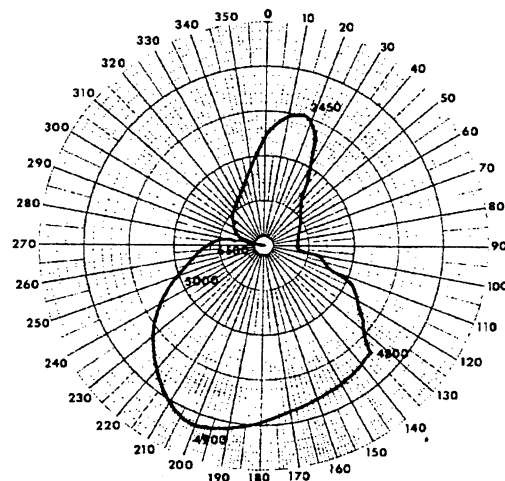
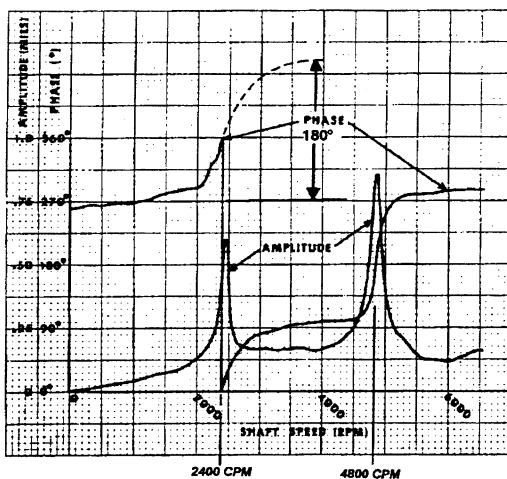


FIGURE 25
COMPARING POLAR AND BODE' PLOTS (Ref. 10)

By looking at the 4900 CPM peak, it is easy to conclude that it is resonant because of the corresponding 90° phase shift in the upper phase-versus-RPM plot (then 180° phase change when it passed through resonant amplification). But is the 2400 CPM peak resonant? If the phase trace at the base of the 2400 CPM peak is transposed to the trace at 360° as shown by the dashed lines in the figure, the 180° phase shift can be seen here also. It is resonant.

As a comparison, notice that orbits in the polar plot are 180° apart (at the unbalance vectors). The resonance response is clearly displayed, although the RPM's had to be manually entered.

4.424 Applying Polar Plots to Natural Frequency and Resonance Diagnostics (Ref. 10)

While polar plots can be used to diagnose problems not caused by or connected with resonance, this discussion will focus on using polar plots in resonance diagnostics only. Polar plots are obtained at each bearing of the machine. Usually two non-contacting pickups are used, placed 90° apart usually in the 9:00 and 12:00 positions, but preferably in the 10:30 and 1:30 positions to be removed from the split in the bearing cup halves (45° off vertical at each location). The polar plots from each bearing are compared to make a final analysis. They will provide "mode shape" information of each resonance, if it is present.

To illustrate, assume that the two plots in Figure 26 were obtained at the two bearings of a large turbine during start-up. Note that both plots reveal amplitude peaks at speeds of 1200, 2200 and 3000 RPM. The amplitude peaks together with the corresponding shifts in phase confirm that all three are resonant frequencies. In addition, it can be noted that both bearings show the same phase indication for the resonant amplitude peak at 1200 RPM. This suggests that the resonance at 1200 RPM is the first rigid mode of the bearings (structure) diagrammed in Figure 27. Rigid modes are associated with structured support natural frequencies which typically occur below the first bending mode of the rotor. Thus, the rigid modes of structure support have lower frequencies than those of the bending modes of the rotor. Further, the plots show that the amplitude peaks at 2200 RPM have opposite (180°) phase. This would indicate that the resonance at 2200 RPM is the second rigid mode of the bearings as illustrated in Figure 27. Finally, since the amplitude peaks at 3000 RPM again show identical phase indications on the two plots, it can be concluded that the resonance at 3000 RPM is the first bending critical of the turbine rotor as illustrated in Figure 27. Although this same information and interpretation could be obtained from Bode' plots, the comparison is made considerably easier with polar amplitude and phase plots.

4.425 Limitations of Polar Plots

Despite advantages and many useful applications, Polar plots also have limitations when compared to amplitude and phase-versus-RPM plots. Major limitations include:

1. While the Polar plot does reveal the presence and significance of resonant conditions during run-up or coastdown, it does not provide a speed reference to indicate the RPM at which resonance occurs. As a result, it is necessary to monitor machine speed and manually record RPM values as noted on the sample plots in Figures 25 and 26.
2. The Polar plot is a plot of the unbalance (synchronous) vibration and, thus, is restricted to examining only the vibration occurring at 1X RPM. As indicated previously, severe resonant conditions can be excited by vibration frequencies other than 1X RPM unbalance. Or, during start-up or coastdown, additional problems such as oil whirl, resonance whirl or rubs may occur resulting in nonsynchronous vibration frequencies. Evidence of these and other problems can be seen when the overall or unfiltered vibration is plotted versus RPM. Such problems are likely to go undetected if only Polar plots are obtained.

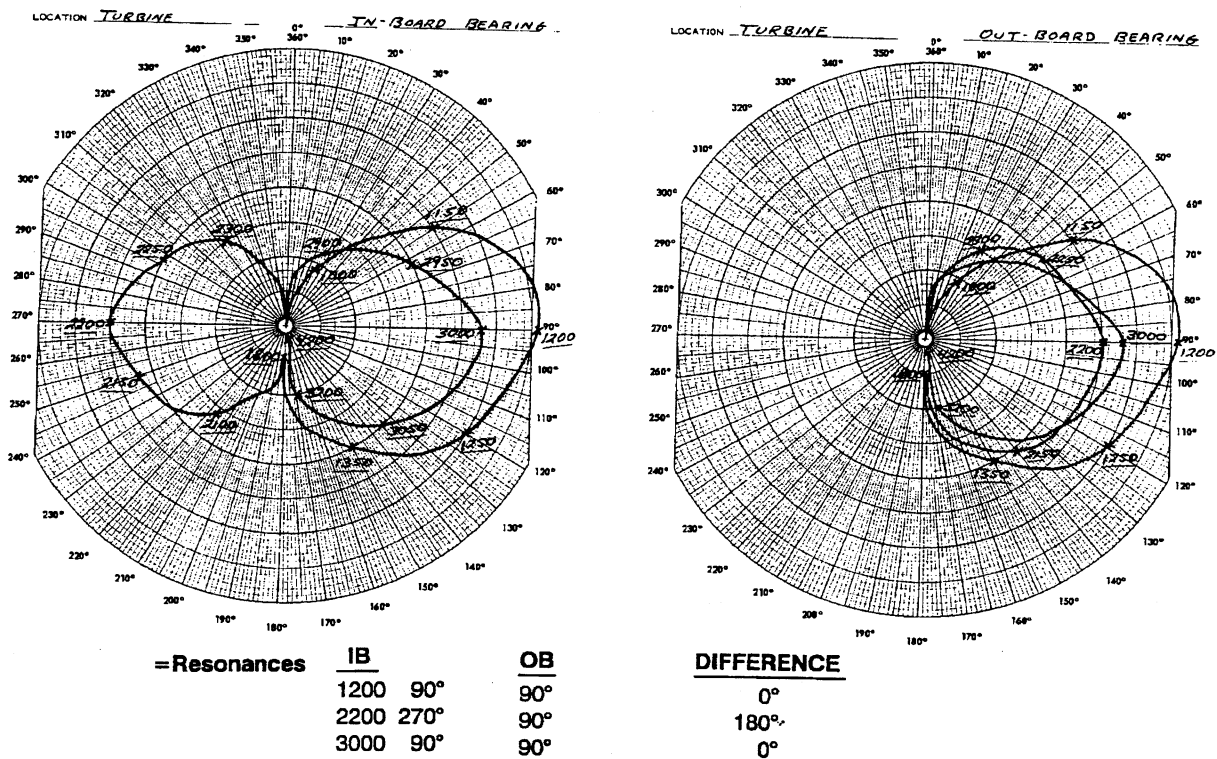


FIGURE 26
COMPARATIVE POLAR PLOTS OBTAINED FOR EACH BEARING OF THE MACHINE CAN AID IN IDENTIFYING RESONANT MODE SHAPES (Ref. 10)

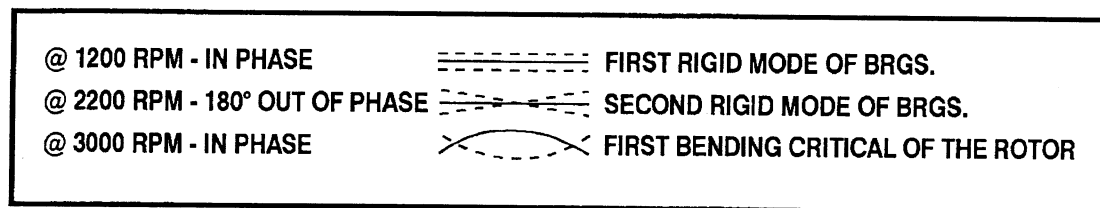


FIGURE 27
RESONANT MODE SHAPES

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